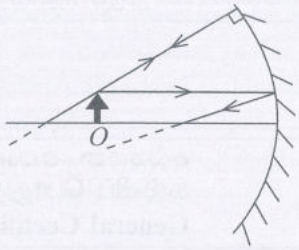
11. The diagram shows the paths of a pair of rays emitted by an object '*O*' placed at a certain point on the principal axis perpendicular to it in front of a concave mirror. The image of this object is,



concave mirror

(1) real, inverted and magnified.
(2) real, inverted and equal to the object in size.
(3) real, upright and reduced.
(4) virtual, upright and magnified.

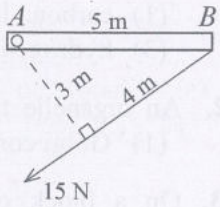
12. A transformer is supplied with a power of 690 W from a supply of 230 V. How much is the current supplied to the coil?

(1) 1 A (2) 2 A (3) 3 A (4) 4 A

13. Which option correctly indicates the nature of radio waves and waves created on a water surface?

	Radio waves	Waves on a water surface
(1)	transverse - mechanical	transverse - electromagnetic
(2)	transverse - electromagnetic	transverse - mechanical
(3)	longitudinal - mechanical	longitudinal - mechanical
(4)	longitudinal - electromagnetic	longitudinal - electromagnetic

14. A force of 15 N is applied by a light string tied to the end *B* of a light rod *AB* which is pivoted at point *A* as shown in the diagram. How much is the moment of force acting on the rod?



(1) (15×5) N clockwise (2) (15×5) N anti-clockwise
(3) (15×3) N clockwise (4) (15×3) N anti-clockwise

15. *X*, *Y* and *Z* are three successive elements in the periodic table and *X* is a noble gas. As which of the following species is *Z* most likely to exist in its compounds?

(1) Z^+ (2) Z^{2+} (3) Z (4) Z^-

16. Four rules relating to electromagnetism and electromagnetic induction are given below.

A - Maxwell's corkscrew rule *B* - Right hand rule
C - Fleming's left hand rule *D* - Fleming's right hand rule

Of these, the two rules that can be used to find the direction of the magnetic field around a conductor carrying an electric current are,

(1) *A* and *B*. (2) *A* and *C*. (3) *B* and *C*. (4) *B* and *D*.

17. Consider the following statements.

A - An unbalanced force does not act on an object at rest freely suspended by a string.
B - An unbalanced force acts on an object moving with a uniform velocity.
C - The force exerted on the table by an object kept at rest on a horizontal table is equal to the force exerted by the table on the object in the opposite direction.

Of the above, the true statement/statements is/are,

(1) only *A*. (2) only *A* and *B*. (3) only *A* and *C*. (4) only *B* and *C*.

18. Which of the following option is true about the muscle tissue shown in the figure?



(1) uninucleate, has cross striations, is not fatigued whatever
(2) multinucleate, no cross striations, is not fatigued whatever
(3) uninucleate, no cross striations, is fatigued on excessive action
(4) uninucleate, has cross striations, is fatigued on excessive action

19. Of the following factors essential for photosynthesis, which factor **cannot** be tested in the school laboratory?

(1) water (2) chlorophyll (3) light (4) carbon dioxide

20. The hydrocarbon with molecular formula C_5H_{12},

(1) is an alkene.
(2) is a component of biogas.
(3) can be polymerised.
(4) forms 6 moles of water on complete combustion of one mole.

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